

ANIKET DESHPANDE

+1 (480) 796-0191 | adeshp61@asu.edu | [linkedin.com/in/aniketdeshpande791](https://www.linkedin.com/in/aniketdeshpande791) | aniketdeshpande.my | github.com/Aniket19j8

PROFESSIONAL SUMMARY

Data Scientist & ML Engineer with proven expertise in end-to-end ML solutions, Generative AI, and cloud ETL pipelines. Delivered a predictive maintenance model for Suzlon Energy (95% accuracy) and led an AI enablement project at Arizona State University that created a production RAG ticket-resolution. Skilled in Python, SQL, and Deep Learning, focused on driving AI-powered product innovation and efficiency.

SKILLS

Languages: Python, SQL, PySpark, R, Java

Machine Learning: Scikit-learn, XGBoost, LightGBM, TensorFlow, Keras, Regression, Classification, Clustering, Time Series, SHAP, Optuna

Generative AI & Agentic Systems: LLMs, RAG Pipelines, LangChain, LangGraph, Agent Orchestration, BERT, Transformers, Prompt Engineering, Semantic Search, Vector Search

Data Engineering: Apache Spark, Airflow, Kafka, Presto, Databricks, ETL, Data Modeling & Mapping, Outlier Treatment

Cloud, MLOps & Deployment: AWS (S3, EMR, SageMaker), Azure ML, Databricks, Docker, Terraform, CI/CD, MLflow, FastAPI, Flask, Model Monitoring, Drift Detection, Experiment Tracking

BI & Visualization: Power BI, Tableau, Matplotlib, Excel

EDUCATION

Arizona State University | Tempe, Arizona, USA

Aug 2024 - May 2026

M.S., Data Science, Analytics & Engineering, GPA 3.8/4.0

Coursework: Machine Learning, Statistical Modeling, Big Data Analytics, Probability, Cloud Computing

MIT World Peace University | Pune, India

Aug 2019 - May 2023

PG Diploma, AI & Machine Learning, GPA 3.7/4.0 | B.C.A., GPA 3.8/4.0

PROFESSIONAL EXPERIENCE

EdPlus at Arizona State University | Instructional Design Assistant

Sep 2024 - May 2026

Scottsdale, Arizona, USA

- Applied UX research methods (structured interviews, usability testing, synthesis) to create standardized QA intake forms, study templates, and Ally accessibility-tagging frameworks; used Tableau to track engagement, completion, accessibility, and performance KPIs, which guided the redesign of 50+ course standards and improved overall learner accessibility and engagement
- Led AI enablement research for Salesforce ticket operations: identified triage and routing friction across support workflows, evaluated GenAI and automation use cases against governance and security requirements, and authored a reusable playbook and reference workflow that became the foundation for a production RAG based ticket resolution system, now used as the team's model for moving AI pilots into production
- Partnered with cross functional teams at project kickoff to scope requirements and success criteria, collaborating with engineers on data pipelines and audit processes to ensure secure, governed access to data for AI enabled workflows
- Earned 4 Sun Devil Excellence Awards from EdPlus leadership for course data review, program readiness checks, and exemplary service in QA and instructional design impact

Suzlon Energy Pvt. Ltd. | Data Analyst, Predictive Analytics

May 2021 - Aug 2021

Pune, India

- Developed and deployed predictive maintenance models (Random Forest, XGBoost) on 500GB+ SCADA time series data using advanced feature engineering, achieving 95% failure detection accuracy and reducing unexpected turbine downtime
- Built real time anomaly detection pipelines (Python, SQL) with statistical outlier treatment and threshold-based alerting, reducing manual diagnostic effort by 25% and visualizing KPIs across 200+ turbines in Tableau dashboards

PROJECTS

Hybrid RAG Ticket Resolution System | AI Research Capstone

2026

- Built an end-to-end RAG pipeline (LangChain, LangGraph, ChromaDB, Pinecone, BM25, cross encoder reranking) combining dense and keyword retrieval for high accuracy semantic search, applying LLMs with prompt engineering and context aware retrieval to automate ticket resolution
- Evaluated the system through stakeholder feedback loops, producing governance documentation and iterative improvements that reduced manual support workload and enabled enterprise grade AI deployment

Fraud Detection with Drift Aware Adaptive ML | Fintech System

2025

- Built a drift aware ensemble fraud detection system (XGBoost, Random Forest, Logistic Regression) on 280K+ transactions using SQL and PySpark, implementing concept drift detection (KS-test, PSI) with automated retraining via Spark and Hive to achieve 83% recall with reduced false positives
- Designed production ready ML monitoring pipelines with MLflow experiment tracking and performance evaluation, and explored a microservices based deployment architecture for model serving endpoints

CERTIFICATIONS & ACHIEVEMENTS

- AWS Certified Machine Learning Specialty, Google Data Analytics Certificate, IBM Data Science Professional Certificate, Applied NLP (Infosys Springboard)
- 4x Sun Devil Excellence Award (EdPlus, ASU), Top 3% Academic Rank across B.C.A. & PGDM AI-ML cohorts (2022, 2023)